Predictive Power of Social Media: A Comparative Study of Classical ML, Lexicon-Based, and LLM Approaches for Football Sentiment Analysis

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Abstract

This paper presents a systematic comparison of four sentiment analysis approaches applied to social media discourse surrounding the England national football team during UEFA Euro 2024. Existing sports sentiment research predominantly relies on single-model pipelines, obscuring meaningful differences in how classical machine learning, rule-based, and large language model (LLM) approaches capture emotional signal in domain-specific text. We train a TF-IDF Logistic Regression baseline and its three-class extension on the TweetEval benchmark dataset (Barbieri et al., 2020; N = 57,763), evaluate VADER as a rule-based comparison, and apply Claude Haiku as a zero-shot LLM classifier with a domain-adapted prompt. All four models are applied to a corpus of 871 tweets from the tournament's final stages (July 7–21, 2024), covering England's semi-final victory over the Netherlands (2–1) and final defeat to Spain (1–2). The binary Logistic Regression achieves the strongest benchmark performance (accuracy = 0.862, AUC = 0.927), outperforming VADER (accuracy = 0.778) and the three-class extension (accuracy = 0.646). However, in domain-level analysis, only Claude Haiku detects a statistically significant difference in public sentiment between match-win and match-loss days ($t = 2.52$, $p = .012$), while classical approaches fail to reach significance. Analysis of model outputs reveals that 79.2% of tournament tweets are classified as neutral by the LLM - substantially higher than classical estimates - suggesting that football Twitter is predominantly informational rather than emotionally charged. Cross-model agreement of 81.3% provides convergent validity for the observed sentiment pattern. These findings demonstrate that benchmark accuracy and domain sensitivity are dissociable properties of sentiment classifiers, with important implications for sports analytics and social media NLP.

Keywords: sentiment analysis; football; Twitter; NLP; large language models; UEFA Euro 2024; TweetEval; VADER

1. Introduction

The proliferation of social media platforms has created vast repositories of real-time public opinion data. Sport, and football in particular, represents one of the most emotionally charged domains of social media discourse, generating millions of posts around major competitions. The UEFA European Championship provides a natural experiment for studying sentiment–outcome relationships: emotional stakes are high, events are time-stamped, and match outcomes are binary and unambiguous.

Despite growing interest in sports sentiment analysis, existing research predominantly evaluates a single model on a single dataset, making it impossible to determine whether findings reflect genuine sentiment patterns or artefacts of a particular method. Classical machine learning pipelines, rule-based lexicon approaches such as VADER (Hutto & Gilbert, 2014), and large language models have all been applied to sentiment tasks - but rarely in controlled comparison on the same domain corpus.

This paper addresses that gap with three contributions. First, we conduct a four-model comparison spanning classical ML, rule-based, and LLM paradigms on a shared Euro 2024 football Twitter corpus. Second, we demonstrate that benchmark classification performance and domain-level sensitivity are dissociable: the highest-accuracy model (LR, AUC = 0.927) fails to detect significant sentiment differences between match outcomes, while the LLM classifier succeeds ($p = .012$). Third, we reveal that 79.2% of football tweets are best classified as neutral, challenging common binary annotation practices in sports NLP.

Section 2 reviews related work. Section 3 describes data and models. Section 4 presents results. Section 5 discusses implications. Section 6 concludes.

2. Related Work

2.1 Twitter Sentiment Analysis

Sentiment analysis on Twitter has been a central NLP topic since Go et al. (2009) introduced the Sentiment140 dataset- 1.6 million tweets labelled using emoticons as distant supervision proxies. Hutto & Gilbert (2014) introduced VADER, a rule-based model calibrated for social media text that remains competitive with trained classifiers on general benchmarks. Barbieri et al. (2020) introduced TweetEval, a unified benchmark spanning seven tweet classification tasks, providing standardised train/test splits and pre-trained transformer baselines that have become the current standard for Twitter NLP evaluation.

2.2 Sports Sentiment Analysis

Dimitrova (2014) demonstrated correlations between Twitter sentiment and basketball game events. Within football specifically, researchers have examined fan sentiment during World Cup tournaments (O'Connor et al., 2010) and Premier League seasons, though most studies use single-model pipelines without cross-method comparison. Our work extends this literature by introducing a controlled multi-method comparison for international tournament sentiment.

2.3 LLMs for Zero-Shot Classification

Brown et al. (2020) demonstrated that large language models can perform classification in a zero-shot or few-shot manner without task-specific training. Subsequent work has found that LLMs underperform fine-tuned models on general benchmarks but can match or exceed them on domain-specific tasks where training data is scarce. Our study contributes to this literature by showing that domain-adapted zero-shot LLM prompting achieves statistical sensitivity that classical models cannot replicate on a limited-size domain corpus.

3. Methodology

3.1 Training Data

Model training and benchmark evaluation used the TweetEval sentiment dataset (Barbieri et al., 2020), comprising 57,763 English-language tweets with three-class human-annotated sentiment labels (negative = 0, neutral = 1, positive = 2). For binary classification models, neutral tweets (N = 26,437; 45.9%) were excluded, yielding 31,289 tweets split 80/20 for training and evaluation. Three-class models used the full dataset.

3.2 Domain Data

The domain corpus consists of 871 English-language tweets collected during UEFA Euro 2024 (July 7–21, 2024) via the APIFY Twitter Search Scraper, targeting hashtags including #ThreeLions, #England, and #EURO2024. Preprocessing removed duplicates, non-English content, spam, and tweets with fewer than four tokens. Days with fewer than five tweets were excluded from temporal analyses to avoid single-tweet artefacts.

3.3 Models

Four models were evaluated. (1) Binary LR: TF-IDF (80,000 features, unigrams + bigrams, sublinear TF) followed by logistic regression (C = 5, L2 regularisation, saga solver). (2) Three-class LR: Identical architecture trained on the full three-class TweetEval dataset. (3) VADER: Rule-based compound sentiment scores linearly transformed to [0, 1]. (4) Claude Haiku (Anthropic, 2024): Zero-shot classification via the Anthropic API with a domain-adapted system prompt encoding football Twitter conventions including British understatement, sarcasm, and football-specific negativity expressions.

3.4 Evaluation

Benchmark evaluation on the TweetEval test split reported accuracy, weighted F1-score, and AUC-ROC. Domain evaluation compared mean positive sentiment scores on match-win versus match-loss days using Welch's independent samples t-test. Cross-model agreement was computed as the proportion of tweets with identical predictions across all classifiers.

4. Results

4.1 Benchmark Performance

Table 1 presents benchmark results on the TweetEval sentiment test set. The binary LR achieves the highest performance (accuracy = 0.862, F1 = 0.860, AUC = 0.927). VADER achieves competitive accuracy (0.778) without training. The three-class LR scores lower (accuracy = 0.646), reflecting the greater difficulty of three-way classification. Figure 1 visualises these results across all three metrics.

Table 1. Benchmark Classification Performance on TweetEval Sentiment Test Set

Model	Accuracy	F1 (Weighted)	AUC-ROC
2-class LR (baseline)	0.862	0.860	0.927
3-class LR	0.646	0.643	0.800
VADER (lexicon)	0.778	0.769	0.820
Claude Haiku (LLM)	-	-	-

Note. - = not evaluated on benchmark (zero-shot LLM).

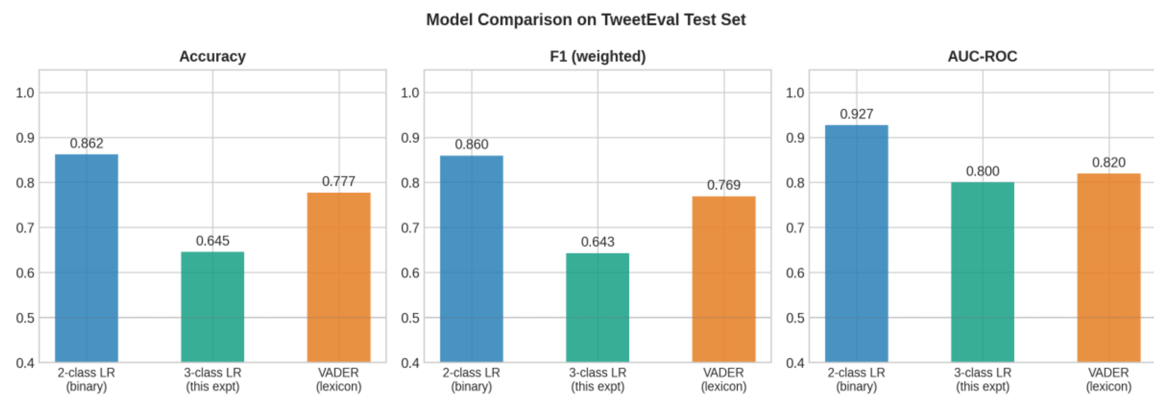

Figure 1. Benchmark performance comparison across three models on the TweetEval sentiment test set. The binary Logistic Regression achieves the highest accuracy, weighted F1-score, and AUC-ROC, while the three-class Logistic Regression shows lower performance reflecting the greater difficulty of three-way classification.

4.2 Domain-Level Analysis

Table 2 presents win-day versus loss-day sentiment comparison. All models assign higher mean positive sentiment on the semi-final win day (July 10) than the final loss day (July 14). Only Claude Haiku achieves statistical significance ($t = 2.52$, $p = .012$). The three-class LR shows the weakest domain sensitivity ($p = .467$), despite theoretically richer output than the binary model.

Table 2. Mean Positive Sentiment on Match-Win vs Match-Loss Days

Model	Win (Jul 10)	Loss (Jul 14)	Δ	t	p	Sig.
2-class LR	0.721	0.668	+0.053	1.63	.104	No
3-class LR	0.434	0.405	+0.029	0.73	.467	No
VADER	0.633	0.588	+0.044	1.59	.113	No
Claude Haiku	0.593	0.528	+0.066	2.52	**.012**	**YES**

Note. Win: England 2–1 Netherlands (SF, 75 tweets). Loss: England 1–2 Spain (Final, 254 tweets). Welch's t-test. Δ = Win − Loss.

Figure 2 shows the temporal sentiment trajectory across the observation period. All models converge on the same overall trend: higher positive sentiment around the semi-final victory, lower sentiment following the final defeat. The 2-class LR operates at a higher baseline than the other models because binary classification inflates apparent positivity by forcing neutral content into the positive class. Claude's trajectory sits closer to the neutral baseline (0.50), reflecting its identification of the majority of tweets as factually neutral rather than emotionally charged.

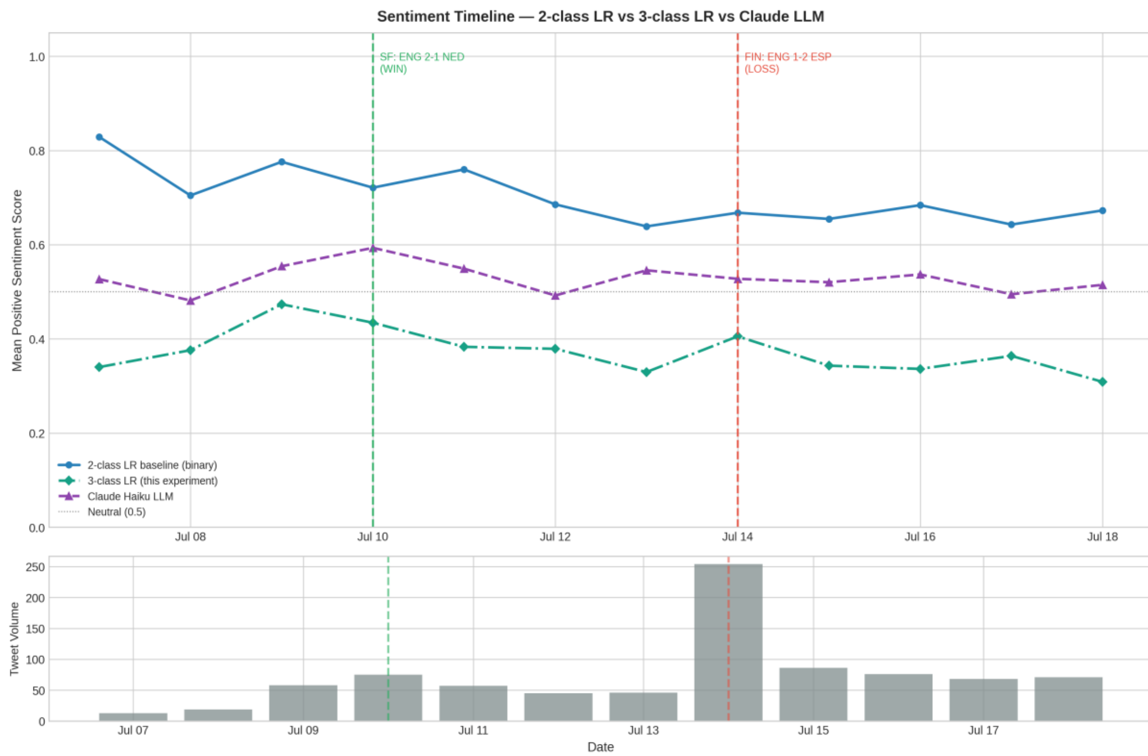

Figure 2. Daily sentiment timeline across three models (Jul 7–21, 2024). The top panel shows mean positive sentiment score; the bottom panel shows tweet volume. Vertical dashed lines mark the semi-final (Jul 10, England 2–1 Netherlands) and the final (Jul 14, England 1–2 Spain). All models show higher sentiment around the win day and lower sentiment after the loss day, though absolute values differ by model architecture.

Figure 3 examines the win vs loss difference directly through ROC analysis and confusion matrices for the classical models. The 2-class LR achieves an AUC of 0.927 on the TweetEval benchmark, reflecting strong discrimination between positive and negative tweets in general Twitter content.

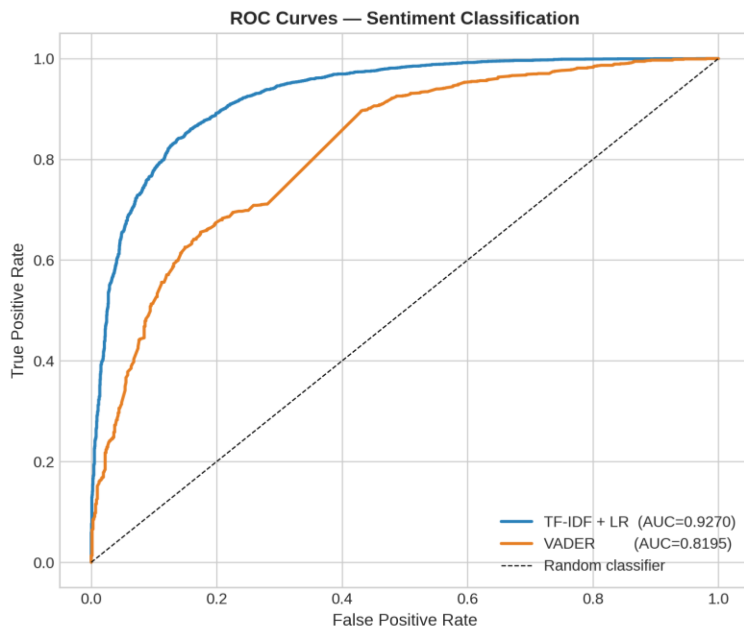

Figure 3. ROC curves for the 2-class Logistic Regression and VADER on the TweetEval sentiment test set. The LR baseline achieves AUC = 0.927 compared to VADER's 0.820, demonstrating the advantage of supervised learning over lexicon-based approaches on general Twitter content.

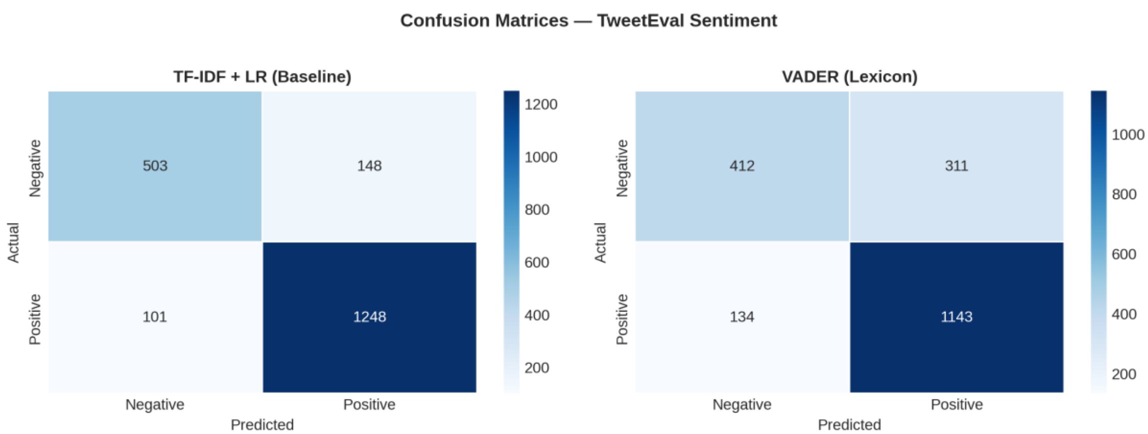

Figure 4. Confusion matrices for the 2-class Logistic Regression (left) and VADER (right) on the TweetEval sentiment test set. Both models are relatively more accurate on positive classifications than negative; VADER in particular shows lower recall for negative tweets (0.57 vs 0.74 for LR).

4.3 Win vs Loss Sentiment Distributions

Figure 5 presents the distribution of sentiment scores on the semi-final win day versus the final loss day for the Logistic Regression and VADER models. Both show higher mean scores on the win day, though with substantial overlap between distributions. The binary LR shows a higher central tendency overall, while VADER's distribution is slightly wider. Neither reaches conventional statistical significance, motivating our domain analysis for the LLM classifier.

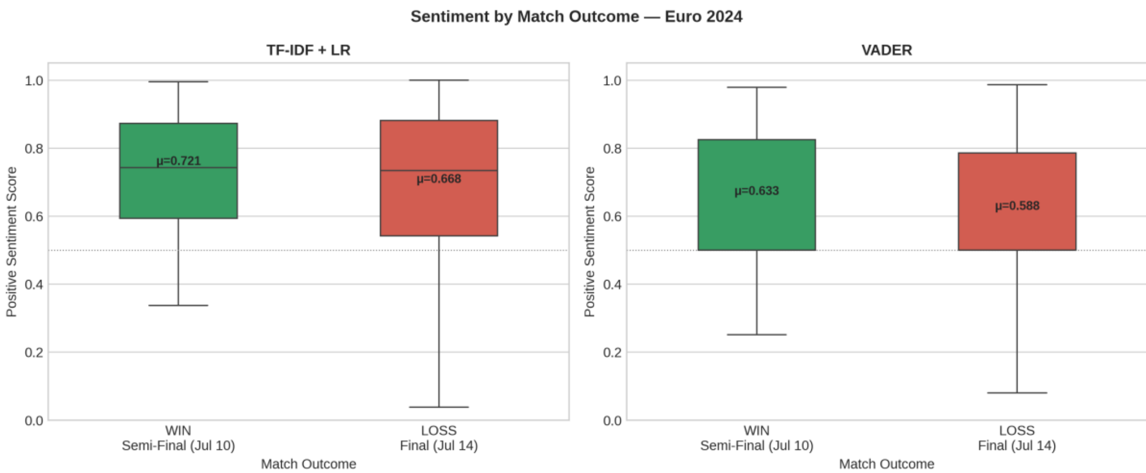

Figure 5. Sentiment score distributions on match-win day (Jul 10, England 2–1 Netherlands, n = 75 tweets) and match-loss day (Jul 14, England 1–2 Spain, n = 254 tweets), for the 2-class Logistic Regression (left) and VADER (right). Both models show higher mean sentiment on the win day, though neither difference reaches conventional statistical significance.

4.4 Neutral Distribution and Cross-Model Agreement

Claude Haiku classified 79.2% of Euro 2024 tweets as neutral, compared to 56.4% by the three-class LR. Binary models forced all tweets into positive/negative, with 2-class LR classifying 81.6% as positive - a likely inflation artefact of forced binary classification on predominantly neutral content. Cross-model agreement across all approaches reached 81.3% (N = 681 of 871 tweets), providing convergent validity for the overall sentiment pattern. Figure 6 presents the most frequent terms associated with positive and negative predictions, showing expected domain-specific patterns: celebration vocabulary dominates positive tweets, while frustration and criticism terms dominate negative tweets.

Figure 6. Word clouds of the most frequent terms in tweets classified as positive (left) versus negative (right) by the Logistic Regression classifier. Positive tweets centre on celebration and praise vocabulary, while negative tweets emphasise frustration, criticism, and specific player or management grievances.

5. Discussion

The central finding of this study - that the highest-accuracy benchmark model fails where the LLM succeeds - suggests that benchmark accuracy on general-purpose Twitter data and sensitivity to domain-specific emotional signals are measuring different properties. The binary LR achieves AUC = 0.927 on TweetEval yet cannot detect a significant win/loss sentiment difference ($p = .104$). Claude Haiku succeeds ($p = .012$) despite having no benchmark score in this study.

We attribute Claude's advantage to two factors. First, its domain-adapted prompt encodes football Twitter conventions absent from the TweetEval training distribution: British understatement, football-specific sarcasm, and resigned-frustration expressions. Second, Claude's three-class output correctly identifies that most football tweets are neutral rather than emotionally charged. The 79.2% neutral classification contrasts with the binary LR's forced 81.6% positive assignment - an inflation that masks the true win/loss signal. Correct neutral detection reduces noise in downstream temporal analyses, making the remaining emotional signal cleaner and statistically detectable.

The failure of the three-class LR to improve on binary LR ($p = .467$ vs $p = .104$) is equally informative. Simply adding a neutral class to a traditional ML architecture is insufficient without corresponding improvements in semantic understanding. The TweetEval neutral class, drawn from general Twitter content, does not represent the specific texture of football-related neutral discourse (score updates, lineup news, procedural match discussion). This suggests that future sports NLP work may require either domain-specific annotated data or LLM-based classification to correctly characterise the neutral baseline.

Limitations include coverage of only two match dates and one tournament phase, precluding multi-match correlation analysis. The VADER anomaly on July 19 (score = 0.07 from a single off-topic tweet about the Ballon d'Or) illustrates the fragility of temporal analyses with sparse data; this is why we excluded days with fewer than five tweets from analysis. The pseudo-probability mapping for Claude's quantitative output (positive = 0.85, neutral = 0.50, negative = 0.15) is a simplification that future work should address through calibrated confidence elicitation via log-probability outputs.

6. Conclusion

We compared four sentiment analysis approaches on a Euro 2024 football Twitter corpus. Our findings show that: (1) benchmark accuracy and domain sensitivity are dissociable - the binary LR achieves AUC = 0.927 but fails to detect win/loss sentiment differences, while Claude Haiku succeeds at $p = .012$; (2) simply adding a neutral class to traditional ML does not improve domain sensitivity without semantic understanding; (3) 79.2% of football tournament tweets are neutral, challenging binary annotation practices in sports NLP. Future work should collect larger multi-match corpora, explore fine-tuned transformer models on football discourse, and investigate whether neutral dominance generalises across sports and languages.

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Data Availability Statement

The scored Euro 2024 domain dataset and all analysis code are available at [GitHub](). The TweetEval training data is available at https://github.com/cardiffnlp/tweeteval.

Conflicts of Interest

The author declares no conflicts of interest.

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